

1) Use the fundamental theorem of calculus and the chain rule to evaluate the following integral.

$$\frac{d}{dx} \int_a^{x^2} \sqrt{t} dt$$

Chain Rule

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) g'(x)$$

$x^2 = u(x)$

$$\frac{d}{dx} \int_a^{u(x)} f(t) dt = f(u(x)) \cdot u'(x)$$

$$f(t) = \sqrt{t} \quad u(x) = x^2 \Rightarrow u'(x) = 2x^1$$

power

$$\frac{d}{dx} \int_a^{x^2} \sqrt{t} dt = \sqrt{x^2} \cdot \frac{d}{dx} (x^2)$$

$$\frac{d}{dx} \int_a^{x^2} \sqrt{t} dt = |x| \cdot 2x = 2x|x|$$

$2x^2$

2) Use the fundamental theorem of calculus and the chain rule to evaluate the following integral.

$$\frac{d}{dx} \int_3^{\sin(x)} t^2 dt$$

Chain Rule

$$\frac{d}{dx} [f(g(x))] = f'(g(x)) g'(x)$$

$$f(t) = t^2 \quad u(x) = \sin(x) \Rightarrow u'(x) = \cos x$$

$$\frac{d}{dx} \int_3^{\sin x} t^2 dt = (\sin x)^2 \cdot \frac{d}{dx} \sin x$$

$$\frac{d}{dx} \int_3^{\sin x} t^2 dt = \sin^2 x \cos x$$

3) A server at a restaurant earns tips at rate of $5t^{\frac{2}{3}}$ dollars per hour worked. To the nearest penny, how much will a server earn after working a 6 hour shift. Tip: do the integration by hand and use a calculator when evaluating the final step. Try to avoid calculators mid computation.

$$\int_0^6 5t^{\frac{2}{3}} dt$$

\$51.43 + min wage

4) On a particularly long drive, a certain car uses gasoline at a rate of $1.8 + 0.2 \cos(x)$ gallons per hour. The cosine term is used to account for road conditions and various starts/stops when driving. Find the average rate of consumption in gallons per hour over the first 4 hours of the trip. Tip: Use a calculator to evaluate at the very end and round to two decimal places of precision. Also recall that the antiderivative of cosine is sine.

$$\int_0^4 0.2 \cos(x) + 1.8$$

$$\int \cos x dx = \sin x \quad \int c dx = cx$$

$$\int (0.2 \cos x + 1.8) dx = 0.2 \sin x + 1.8x$$

$$\left[0.2 \sin x + 1.8x \right]_0^4 = (0.2 \sin(4) + 1.8(4)) - (0.2 \sin(0) + 1.8(0))$$

$\sin 0 = 0$

$$0.2 \sin 4 + 7.2$$

$$-0.7568 + 7.2$$

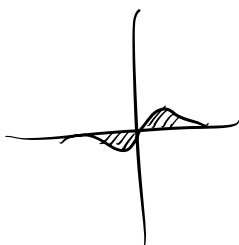
$$7.0486$$

7.05 gallons over 4 hrs

5) Evaluate the following integral by plotting the function (preferably on desmos) and analyzing its symmetry.

$$\int_{-4}^4 x e^{-x^2} dx$$

Desmos sketch



Symmetric around zero

the Area of the integral
When playing with the bounds, v approaches
zero as $|lower\ bound| \rightarrow |upper\ bound|$

6) Evaluate the following integral using u -substitution. You may wish to plug in the bounds of the previous problem to confirm the answer to problem (5) by hand, rather than relying on symmetry.

$$\int x e^{-x^2} dx$$

$$x e^{-x^2} dx$$

u sub

$$f(x) = -x^2$$

const mult

$$-f(x) = x^2$$

power rule

$$-f'(x) = 2x$$

$$f'(x) = -2x$$

$$\text{let } u = -x^2$$

$$\frac{du}{dx} = -2x \Rightarrow du = -2x dx$$

$$x dx = -\frac{1}{2} du$$

$$\int x e^{-x^2} dx = \int e^u \left(-\frac{1}{2} du\right) = -\frac{1}{2} \int e^u du = -\frac{1}{2} e^u + C$$

$$\int x e^{-x^2} dx = -\frac{1}{2} e^{-x^2}$$

7) Using a picture and a couple of sentences explain why the following integrals must be equivalent. It might help to plot the graph of $\cos(x)$ on desmos to get started.

$$\int_{-2}^2 3 \cos x \, dx = 6 \int_0^2 \cos x \, dx$$

const mult

$$\int c \cdot f(x) \, dx = c \int f(x) \, dx$$

$$3 \int_{-2}^2 \cos x \, dx = 6 \int_0^2 \cos x \, dx$$

$$3 \int_{-2}^2 \cos x \, dx = 3 \int_0^2 \cos x \, dx + 3 \int_{-2}^0 \cos x \, dx = 6 \int_0^2 \cos x \, dx$$

$$\int_{-2}^0 \cos x \, dx = \int_{-2}^0 \cos(-u) \, -du$$

cos even

$$\cos -u = \cos u$$

$$-\int_{-2}^0 \cos(u) \, du = \int_0^2 \cos(u) \, du$$

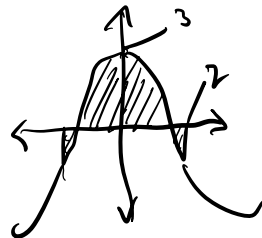
$$3 \int_0^2 \cos x \, dx + 3 \int_0^2 \cos x \, dx$$

$$3(2 \int_0^2 \cos x \, dx) = 6 \int_0^2 \cos x \, dx$$

$$6 \int_0^2 \cos x \, dx = 6 \int_0^2 \cos x \, dx$$

$$\int_{-2}^2 3 \cos(x) \, dx$$

$$6 \int_0^2 \cos(x) \, dx$$



8) Evaluate the following indefinite integral using u -substitution.

$$\int \frac{13x^2}{1-x^3} \, dx$$

$$\int \frac{13x^2}{1-x^3} \, dx$$

$$v = 1-x^3$$

$$dv = -3x^2 \, dx \Rightarrow x^2 \, dx = -\frac{1}{3} \, dv$$

$$\int \frac{13x^2}{1-x^3} \, dx = 13 \int \frac{1}{v} \left(-\frac{1}{3} \, dv\right)$$

$$= -\frac{13}{3} \int \frac{1}{v} \, dv$$

$$\int \frac{1}{v} \, dv = \ln |v| + C$$

$$= -\frac{13}{3} \ln |1-x^3| + C$$

9) Evaluate the following indefinite integral using u -substitution. Hint: The correct choice for u will trivialize this problem.

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

$$\int 12x^3 \sin(3x^4) e^{\cos(3x^4)} dx$$

chain rule $u = \cos 3x^4$

$$\frac{du}{dx} = -\sin 3x^4 \cdot \frac{d}{dx} 3x^4 = -\sin(3x^4) \cdot 12x^3$$

$$12x^3 \sin(3x^4) dx = -du$$

$$\int e^{\cos(3x^4)} \cdot (12x^3 \sin(3x^4) dx) = \int e^u (-du) = -\int e^u du$$

$$-\int e^u du = -e^u + C$$

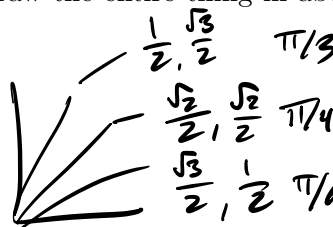
$$-e^{\cos 3x^4} + C$$

$$\frac{\cos(3x^4)}{\sin(3x^4)} \cdot 12x^3$$

10) Evaluate the following integral. Provide an exact answer. You may need to brush up on your unit circle. If anyone needs some pointers on that, I can show you how to draw the entire thing in about 10 minutes via office hours.

$$\int_0^{\sqrt{3}/2} \frac{dx}{\sqrt{1-x^2}}$$

$$\int_0^{\sqrt{3}/2} \frac{dx}{\sqrt{1-x^2}}$$



$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

$$(\arcsin x) \Big|_0^{\sqrt{3}/2} = \arcsin \frac{\sqrt{3}}{2} - \arcsin(0)$$

$$\frac{\pi}{3} - 0$$

$$\frac{\pi}{3}$$